# SSME-Lite: Reproducible Model Evaluation with Scarce Labels

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## Abstract

SSME-Lite combines semi-supervised evaluation with reproducible comparisons and offline diagnostics. PneumoniaMNIST is the main case study: checkpoint inference through report generation, plus a frozen-ResNet18 transfer example. CivilComments, used in the original SSME paper, provides a quickstart and reproduction check. With 20 visible labels, Accuracy estimation error falls from 5.70 to 2.22 percentage points for pretrained checkpoints and from 7.11 to 4.16 for transfer classifiers. SSME-Lite is available on PyPI and connects prediction, evaluation, and reporting through a unified Python interface.

**Keywords:** semi-supervised evaluation, model selection, reproducible software, Python

## 1 Motivation and Contribution

Semi-Supervised Model Evaluation (SSME) estimates the joint distribution of labels and classifier probabilities to evaluate models with scarce labels (Shanmugam et al., 2025). This project provides a reusable estimator, reproducible experiments, and metric, ranking, and sensitivity diagnostics (Figure 1). PneumoniaMNIST demonstrates inference-to-report integration; CivilComments checks reproduction on the paper's data. The contribution is their integration into a documented, reusable Python workflow.




Figure 1: End-to-end evaluation: PneumoniaMNIST starts with checkpoint inference; the CivilComments quickstart uses stored scores. Hidden labels enter benchmark scoring only.

## 2 Semi-Supervised Estimator

Let $p_{imk}$ denote class probabilities, with shape $N \times M \times K$ (binary: $N \times M$). Missing labels use $-1$; each class needs a visible label. Clipped, normalized scores undergo the additive log-ratio transform (Aitchison, 1982) $z_{imk} = \log(p_{imk}/p_{imK})$, yielding $M(K-1)$ features.

With responsibilities $\gamma_{ik}$ and annotation weights $w_i = 10$ for visible labels and 1 otherwise, iterations update Gaussian kernel densities (Scott, 1992), priors, and posteriors:

$$f_k(z) = \frac{\sum_i w_i \gamma_{ik} K_h(z - z_i)}{\sum_i w_i \gamma_{ik}}, \quad \pi_k = \frac{\sum_i w_i \gamma_{ik}}{\sum_i w_i}, \quad \gamma_{ik} \leftarrow \frac{\pi_k f_k(z_i)}{\sum_c \pi_c f_c(z_i)}. \tag{1}$$

Initialization samples from mean model probabilities; visible labels remain clamped. Fixed iterations and explicit bandwidth settings support controlled, reproducible comparisons.

## 3 Implementation and Experimental Design

**Reusable interface and reports.** The prediction adapter aligns class columns; the split utility records sample indices. A scikit-learn-style estimator (Pedregosa et al., 2011) exports offline HTML, CSV metrics, and JSON statistics:

```
from ssme_lite import SSMEEstimator
e = SSMEEstimator(max_iter=100, random_state=0)
e.fit(scores, y_partial, model_names=model_names)
e.report(n_draws=100, contribution=True).save("evaluation")
```

MIT-licensed version 0.1.0 is published on PyPI (`pip install ssme-lite`); the core needs no PyTorch. CI passes on Python 3.10–3.14; local validation passes 38 tests, with one optional reference-code audit skipped.

**Metrics and uncertainty.** The default report evaluates the estimation pool (visible plus unlabeled samples). Accuracy is the exact posterior expectation of a model's correctness. ROC AUC, average precision (AP), and expected calibration error (ECE) average 100 shared latent-label draws, preserving paired comparisons. AP is the implementation's AUPRC summary; binary ECE uses ten equal-width bins (Guo et al., 2017). The interface supports multiclass inputs; the present experiments are binary. Label-sampling intervals describe uncertainty conditional on the fitted posterior.

| Pipeline | Models | $n_L$ / $n_U$ / $n_H$ | Splits | Bandwidth / epochs |
|---|---|---|---|---|
| Checkpoint inference | 7 | 20 / 400 / 204 | 10 | Scott / 100 |
| Transfer learning | 5 | 20 / 400 / 204 | 10 | Scott / 100 |
| CivilComments quickstart | 7 | 20 / 1,000 / 66,891 | 50 | Reference / 20 |

Table 1: The first two pipelines use PneumoniaMNIST, the main dataset. Counts are visible ($L$), unlabeled ($U$), and held-out ($H$); all runs use weight 10 and 100 label draws.

**PneumoniaMNIST: the main end-to-end study.** The dataset has 4,708 training, 524 validation, and 624 test images (Yang et al., 2023). The demo loads seven pretrained checkpoints (one ResNet18, three AutoKeras (Jin et al., 2023), three AutoML Vision), runs test-image inference, aligns probabilities, fits SSME, and exports the interactive report. The transfer example freezes an ImageNet ResNet18 (He et al., 2016) for 512-dimensional features and trains five sklearn heads on training data only, with fixed hyperparameters. Both pipelines evaluate test data over seeds 0–9; validation is unused for transfer-head selection.

**CivilComments: quickstart and reproduction check.** CivilComments (Borkan et al., 2019), used in the SSME paper, supplies seven released score vectors per comment. A half-split protocol gives 50 valid runs (seeds 0–58; nine missing-class draws excluded). The reference-code bandwidth rule is the minimum normal-reference ALR bandwidth divided by four. Archived official-code executions establish the reference workflow; this package-level study checks its qualitative gains.

**Paired error analysis.** Split-level error is $e_s = M^{-1} \sum_m |\widehat{q}_{sm} - q^H_{sm}|$ against held-out truth. Both methods use the same visible labels. Reported reductions $d_s = e^L_s - e^{\mathrm{SSME}}_s$ use descriptive intervals $\bar{d} \pm t_{.975,S-1} s_d/\sqrt{S}$; splits, not individual models, are repetitions.

## 4 PneumoniaMNIST: Main Results and Diagnostics

| Metric | MAE (pp) | | Paired reduction (pp) | |
|---|---|---|---|---|
| | Labels only | SSME-Lite | Mean | 95% CI |
| *Pretrained checkpoints (7 models)* | | | | |
| Accuracy | 5.70 | 2.22 | 3.49 | [2.37, 4.60] |
| AUC | 3.17 | 2.55 | 0.62 | [0.19, 1.04] |
| AP | 2.98 | 2.40 | 0.58 | [-0.06, 1.22] |
| ECE | 3.93 | 2.39 | 1.54 | [0.69, 2.38] |
| *Transfer learning (5 heads)* | | | | |
| Accuracy | 7.11 | 4.16 | 2.95 | [1.00, 4.89] |
| AUC | 4.75 | 3.89 | 0.87 | [-0.59, 2.33] |
| AP | 4.27 | 4.05 | 0.22 | [-1.47, 1.90] |
| ECE | 6.43 | 3.96 | 2.47 | [0.68, 4.26] |

Table 2: PneumoniaMNIST held-out MAE (pp), ten paired splits. Positive reduction favors SSME-Lite; brackets give unadjusted 95% $t$ intervals for the paired reduction.

**Accuracy and calibration show the clearest gains.** Accuracy MAE falls by 61.1% for pretrained checkpoints and 41.4% for transfer heads. All twelve models improve in mean Accuracy MAE (Figure 2). Both pipelines also improve ECE, extending the gains to calibration assessment.

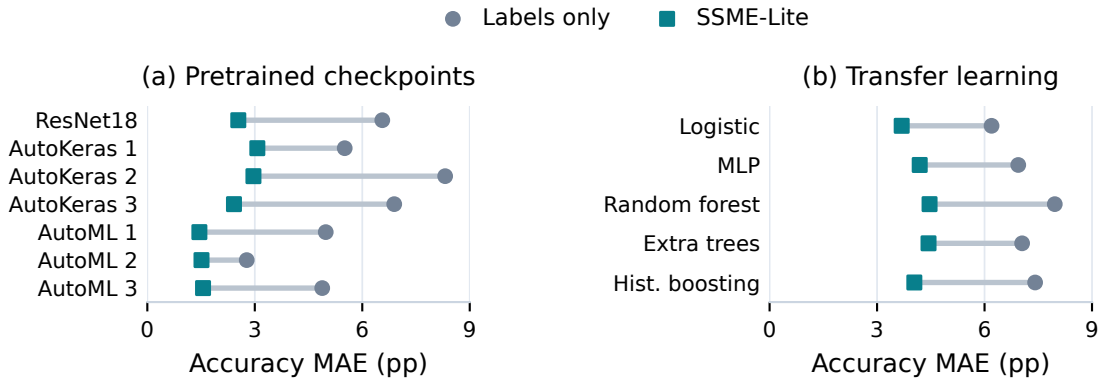



Figure 2: Accuracy estimation MAE for all seven checkpoints and five transfer heads, averaged over ten held-out splits. Lower is better; every model improves on average.

**Improved Accuracy-based model selection.** Accuracy Top-1 success rises from 23.3% to 50.0% for checkpoints and from 25.0% to 70.0% for transfer heads. The workflow thus supports both accurate metric estimation and selection of the highest-Accuracy model.

**Interpretable model-influence diagnostics.** For each removed model $m$, sensitivity is $\mathrm{TV}_m = |U|^{-1} \sum_{i \in U} \frac{1}{2} \|\gamma_i - \gamma_i^{(-m)}\|_1$. Removing MLP or logistic regression gives the largest mean TV (0.0465 and 0.0425); the two forest models yield 0.0043 and 0.0030. Initialization, bandwidth, labels, and epochs are fixed. TV quantifies conditional influence under shared full-view initialization.

## 5 Quickstart Validation and Practical Reuse

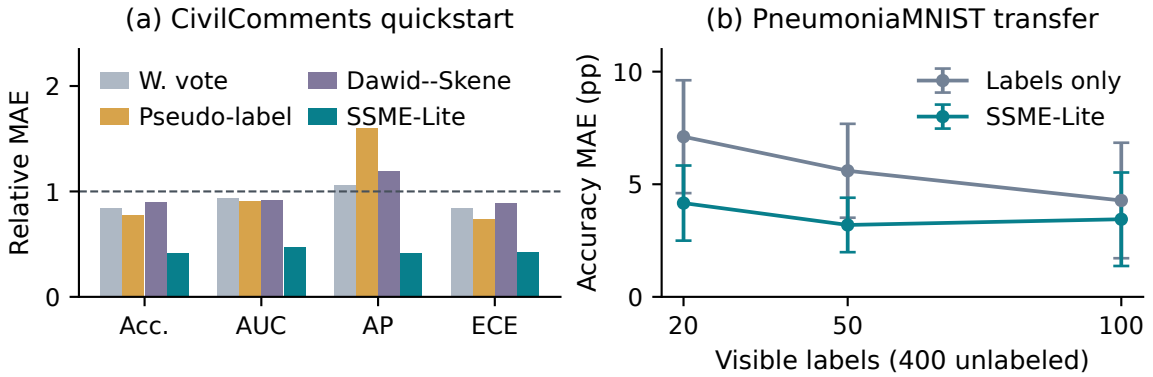



Figure 3: (a) Paper-data quickstart: CivilComments relative MAE over 50 valid splits; dashed line: labeled-only baseline. (b) Main-dataset extension: PneumoniaMNIST transfer MAE over ten splits per budget, with 95% $t$ intervals.

**Quickstart validation on the original paper's data.** CivilComments MAEs for Accuracy, AUC, AP, and ECE are 2.27, 3.28, 9.41, and 2.12 pp; relative errors are 0.41, 0.47, 0.42, and 0.42. The quickstart recovers the advantage over labeled-only estimation, weighted voting, pseudo-labeling, and Dawid–Skene aggregation (Dawid and Skene, 1979) (Figure 3a). This confirms the expected qualitative gains; the package uses equal-width ECE bins, while the paper uses equal-frequency bins.

**Gains across annotation budgets.** With five transfer heads and 400 unlabeled examples, increasing visible labels from 20 to 50 to 100 changes Accuracy MAE from 7.11/4.16 to 5.60/3.19 to 4.28/3.45 pp (labeled-only/SSME). SSME-Lite lowers mean error at every budget (Figure 3b), with the largest relative gain at 20 labels. Holdout sizes are 204, 174, and 124, respectively.

**Reusable outputs and evaluation scope.** Offline HTML, CSV, and JSON support inspection and reuse. A JSON-configured CLI evaluates stored predictions. Parallel prediction, class-density fits, and benchmark seeds complement batched posterior evaluation. Gains are clearest for Accuracy and ECE; AUC/AP selection shows no consistent improvement. Cross-split and conditional label intervals quantify distinct sources of variation.

**Reproducibility and verification.** Source: github.com/jasonshen2002/ssme-lite. Saved runs retain splits, configurations, and versions. The builder checks model-level errors, regenerates figures, and records input hashes. Appendix A gives executable reproduction steps. All ten sensitivity runs are included, enabling verification from saved outputs.

## 6 Conclusion

SSME-Lite delivers a reusable Python workflow, validated through CivilComments reproduction and a PneumoniaMNIST end-to-end and transfer-learning study. Improved Accuracy and ECE estimation, Accuracy-based model selection, and interpretable offline reports demonstrate its practical value for scarce-label evaluation.

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## Appendix A. Experimental Artifacts and Reproduction

**Artifact access.** The released package is `ssme-lite==0.1.0`; source and English documentation are available at github.com/JasonShen2002/ssme-lite. OpenI experiment project: openi.pcl.ac.cn/jasonshen/SSME-Lite-2026211240. The steps below run from the root of the accompanying source archive.

**Installation.** Use a clean Python 3.11 environment. On macOS/Linux:

```
python -m venv .venv
source .venv/bin/activate
python -m pip install -r reproduction/requirements.txt
```

On Windows, activate with .venv/Scripts/Activate.ps1. Verification receipts record the tested environment; fixed-score fits need no GPU.

**Main results: fresh fits from fixed predictions.** Two small NPZ artifacts retain the 624 official test rows, ordered candidate names, and benchmark labels for the seven-checkpoint and five-head studies. Their SHA-256 hashes and provenance are in reproduction/inputs/manifest.json. Run:

```
python reproduction/run.py --jobs 2
```

This performs ten splits per setting: checkpoints with 20 labels, and transfer heads with 20, 50, and 100 labels. Each fit uses 400 unlabeled rows, 100 fixed EM iterations, and 100 posterior label draws. At 20 labels, the runner also refits each leave-one-model-out diagnostic. It checks split indices, model-level metrics, errors, ranking statistics, and posterior-TV values against the supplied references with absolute tolerance $10^{-10}$. New outputs and `verification.json` are written under `results/reproduction/`; the original references are preserved. The expected Accuracy MAEs (labeled-only/SSME, pp) are 5.70/2.22 for checkpoints and 7.11/4.16, 5.60/3.19, and 4.28/3.45 for the three transfer budgets. This route recomputes SSME rather than merely redrawing saved summaries.

**CivilComments quickstart reproduction.** Download the seven released prediction/label pairs following the official SSME repository. Keep each model directory under inputs/CivilComments/, with `preds.npy` and `labels.npy`. Install `ssme-lite[official]==0.1.0`, then run the command below, replacing the input directory with its actual location:

```
python experiments/civilcomments/reproduce_official_split.py --inputs
    /path/to/SSME/inputs --output results/reproduction/civilcomments --jobs 2
```

The script collects 50 valid splits in seed order, using 20 visible labels, 1,000 unlabeled rows, a 66,891-row holdout, reference-code bandwidth, and 20 EM iterations. The expected Accuracy/AUC/AP/ECE MAEs are 2.27/3.28/9.41/2.12 pp.

**Optional image-to-report workflow.** Install `ssme-lite[transfer]==0.1.0`; also install `ai-edge-litert` for checkpoints. Run `pneumonia_ssme.py` or `transfer_ssme.py` in the corresponding experiment directory. They download assets, regenerate predictions, and produce reports. Run the matching experiment's `config.json` with `ssme-lite` for repeated evaluation. See docs/reproduction.md for complete commands and platform-dependent inference/training considerations.